RAMDEOBABA UNIVERSITY, NAGPUR

Department of Computer Science and Engineering

E-Commerce and ERP • TA-2 Theory Assessment

COURSE PROJECT & CASE STUDY REPORT

FreshMart — Online Grocery Shopping Portal

"Fresh Groceries, Happy Lives"

STUDENT CREDENTIALS

Student Name:	`[Insert your full name]`
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Section:	`[Insert your section]`
Degree / Branch:	**B.Tech • CSE**
Academic Year:	**2026–2027**

COURSE & PROJECT INFO

Subject:	**E-Commerce and ERP**
Assessment:	**TA-2 Theory Assessment**
Application Type:	**Single-Page E-Commerce Web App**
Technology:	**HTML5 • CSS3 • Vanilla JS (ES6+)**
Submission:	**Academic Term Evaluation**

Institution: Ramdeobaba University, Ramdeo Tekdi, Gittikhadan, Katol Road, Nagpur, Maharashtra 440013

A comprehensive project report submitted in partial fulfillment of the continuous assessment requirements for E-Commerce and ERP.

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Executive Summary

FreshMart is a single-page web application (SPA) built to model modern quick-commerce grocery operations. Developed with pure HTML5, CSS3, and modern vanilla JavaScript, the system demonstrates the practical convergence of front-facing E-Commerce catalogs with back-office ERP logic, including category item indexing, price and discount computations, dynamic delivery fee thresholds, cart persistence, and customer verification guards.

The project highlights two critical business constraints:

- **Mandatory Authentication Guard:** Customers can freely browse the catalog, inspect product specifications, and filter categories without friction. However, attempts to add items to the shopping cart require an explicit demo login with a 10-digit Indian mobile number. To prevent accidental checkout additions, the system intentionally requires the customer to click "Add to Cart" again after authentication.
- **Consistent Quick-Commerce Delivery Messaging:** Quick-commerce portals differentiate through fast delivery. FreshMart enforces an explicit estimate of **10–20 Minutes** across all customer-facing touchpoints (announcement bar, hero section, product modals, cart drawer, checkout dialog, and order fulfillment screen).

> **Key Deliverable Notice:** The implementation runs entirely locally in standard web browsers, utilizes 33 embedded SVG product illustrations, stores cart and session states in `localStorage`, and requires zero third-party dependencies or external backend servers.

1. Introduction & Background

Electronic Commerce (E-Commerce) refers to the buying and selling of goods and services over electronic networks, primarily the Internet. In modern retail supply chains, digital grocery portals have transformed traditional supermarket shopping into an automated, highly efficient consumer journey. By combining intuitive web interfaces with backend enterprise resource planning (ERP) concepts—such as inventory catalogue classification, dynamic price calculations, delivery rule engines, and order lifecycle management—E-Commerce portals empower consumers to procure daily essentials from the convenience of their homes while reducing operational overhead for retailers.

FreshMart — Online Grocery Shopping Portal (tagline: *"Fresh Groceries, Happy Lives"*) is an educational, production-ready frontend E-Commerce web application developed to demonstrate core principles taught in the **E-Commerce and ERP** curriculum. The portal simulates a modern quick-commerce grocery business model, offering consumers a catalogue of fresh fruits, vegetables, dairy, bakery items, beverages, and pantry staples.

FreshMart addresses common usability and transactional challenges in quick-commerce web design. Specifically, it incorporates:

- **Structured Catalogue Navigation:** Dynamic categorization with real-time product count badges across multiple grocery departments.
- **Instant Product Discovery:** Client-side multi-parameter search combined with category filtering and real-time empty-state feedback.
- **Controlled Transaction Funnel:** A mandatory authentication guard requiring customers to log in before adding items to their cart, avoiding anonymous ghost carts while maintaining frictionless browsing.
- **Automated ERP Pricing Engine:** Instant subtotal, delivery fee calculation (free delivery above ₹500, else ₹40), and dynamic discounts.
- **Hyper-Local Delivery Messaging:** Consistent communication of a **10–20 minute delivery window** across 11 key touchpoints.
- **Persistent State Management:** Full data persistence across browser reloads using HTML5 `localStorage`.

2. Project Objectives

The primary engineering and educational objectives of the FreshMart project include:

1. **Designing a Modern E-Commerce UI:** Build an appealing, responsive grocery shopping portal utilizing an emerald green palette (`#168534`) that aligns with fresh produce retail standards.
2. **Multi-Category Inventory Management:** Structure products into intuitive categories (Fruits & Vegetables, Dairy & Eggs, Bakery & Bread, Beverages, Snacks & Munchies, Pantry Staples) with dynamic counts.
3. **Search & Filter Implementation:** Implement responsive keyword filtering across product names, descriptions, and categories with instantaneous DOM re-rendering.
4. **Enforcing Authentication Gate:** Require customer verification (10-digit Indian mobile number) prior to adding items to the cart, preventing unverified cart modifications while preserving open browsing.
5. **ERP Pricing and Cart Calculation:** Dynamically compute item subtotals, apply delivery fee thresholds (₹40 under ₹500, free at/above ₹500), and display live quantity badges.
6. **Comprehensive Checkout Form Validation:** Validate recipient full name, valid 10-digit phone, street address, pin code, and payment method selection (Cash on Delivery, UPI, Cards).
7. **Simulating Quick-Commerce Fulfillment:** Generate unique order identifiers, clear active carts, persist order summaries to `localStorage` , and confirm delivery within 10–20 minutes.

3. Project Scope

The project scope was defined to balance comprehensive frontend feature richness with self-contained, zero-dependency local execution:

Dimension	In-Scope Features (Implemented)	Out-of-Scope (Future Enhancements)
Frontend Architecture	Pure HTML5, CSS3, Vanilla JavaScript (ES6+); modular CSS custom properties; responsive Flexbox/Grid.	Server-side rendering (SSR), Angular, React, Vue, or TypeScript frameworks.
User Authentication	Client-side 10-digit Indian mobile validation, demo credentials, header profile sync, persistent session.	External SMS gateway integration (Twilio/Fast2SMS), OAuth2/Google Sign-In, biometric auth.
Inventory & Pricing	25 structured grocery products, pricing in INR (₹), discount tags, stock status, delivery fee tier.	Live inventory depletion against relational database (MySQL/PostgreSQL), multi-warehouse ERP.
Payment & Fulfillment	Simulated checkout with COD, UPI QR, and Card radio options; unique order ID generation; order modal.	Production payment gateways (Razorpay, Stripe, PayU), real-time bank settlement, SMS webhook.
Data Persistence	Browser-native `localStorage` for cart items, user session state, and order history logs.	Enterprise cloud database (MongoDB Atlas, AWS DynamoDB), distributed session stores (Redis).

4. Technologies Used & Architectural Rationale

Technology	Component / Role	Architectural Rationale
HTML5	Semantic Page Skeleton	Provides accessible landmarks (`<header>` , `<main>` , `<section>` , `<dialog>`), enabling search crawlers and screen readers to parse the shopping portal natively.
CSS3	Styling & Responsive Layout	Utilizes CSS Custom Properties (variables) for theme consistency (`--primary: #168534`), CSS Grid for catalog layouts, Flexbox for header alignment, and media queries for desktop/tablet/mobile responsiveness.
Vanilla JavaScript (ES6+)	State & DOM Controller	Implements event listeners, array manipulations (`map` , `filter` , `reduce`), modal controls, dynamic HTML generation, and transactional validations without third-party library overhead.
HTML5 Web Storage	Client-Side Persistence	`localStorage` retains user login credentials, cart items, quantities, and past order records across browser reloads without requiring an active backend server.
Scalable Vector Graphics (SVG)	Visual Assets & Produce Imagery	33 custom SVGs provide lightweight, infinitely scalable, high-DPI product artwork and UI icons with zero network latency and complete offline reliability.

> **Stack Independence Rationale:** Relying exclusively on standard web technologies guarantees that FreshMart executes smoothly in any modern browser on Windows, macOS, Linux, and Android without requiring Node.js runtime environments, npm module compilation, or active internet connectivity.

5. System Workflow & Customer Journey

FreshMart models an end-to-end quick-commerce shopping journey. The state machine strictly enforces customer authentication before allowing items to enter the cart, preventing unverified transactions.



6. Major Modules and Features (Part 1)

Module 1: Top Navigation & Global Header

The global header provides a persistent anchor for navigation across the entire site. It incorporates an SVG leaf brand icon, the "FreshMart" wordmark, an interactive delivery pin showing "Deliver to: **Nagpur 440013 (10–20 Mins)**", an omni-channel search bar, an authentication status pill ("Login" or "Hi, +91... / Logout"), and an animated floating cart button with real-time numeric quantity counter badge.

Module 2: Hero Promotional Banner & Service Benefits

Positioned above the fold, the hero banner presents FreshMart's core consumer value propositions: "Fresh Groceries Delivered to Your Doorstep in **10–20 Minutes**". A secondary benefits bar highlights four operational guarantees: *10–20 Min Express Delivery*, *Best Quality Guarantee*, *Free Delivery Above ₹500*, and *100% Secure Payments*.

Module 3: Category Filtering Engine

The left navigation bar and interactive category grid present six grocery departments: *All Categories*, *Fruits & Vegetables*, *Dairy & Eggs*, *Bakery & Bread*, *Beverages*, *Snacks & Munchies*, and *Pantry Staples*. Each category dynamically calculates and displays the current number of available products (e.g., "6 items"), giving shoppers immediate visibility of inventory depth.

6. Major Modules and Features (Part 2)

Module 4: Dynamic Product Catalog & Card Architecture

The main view renders responsive product cards containing high-DPI product artwork, category badges, product titles, unit specifications (e.g., "500g", "1L", "6 pcs"), dynamic price displays in INR (₹) with strike-through MRP comparisons, discount percentage badges (e.g., "15% OFF"), and a responsive "Add to Cart" button.

Module 5: Real-Time Multi-Parameter Search

The search input listens to keyboard input and executes instant client-side filtering across product titles, descriptions, and categories. When no items match the query, the interface cleanly transitions to an empty-state banner displaying an illustrative search icon and a "Clear Search" reset button.

Module 6: Product Quick-View Modal

Clicking on any product card opens an accessible modal dialog revealing high-resolution artwork, detailed nutritional descriptions, shelf-life indicators, storage instructions, unit price, quantity increment/decrement controls, and a prominent "Add to Cart" action button with login status validation.

Module 7: Authentication Guard & Demo Login Subsystem

In accordance with business specifications, unauthenticated users cannot modify the shopping cart. Clicking "Add to Cart" when logged out triggers a non-blocking toast warning: *"Please log in to add products to your cart."* A clean modal prompts the user for their 10-digit Indian mobile number with regex validation (`/^[6-9]\d{9}$/`). The product is never automatically added upon login; the customer explicitly reviews the product and clicks "Add to Cart" again.

Module 8: Shopping Cart & ERP Calculation Drawer

The slide-over cart drawer maintains synchronisation with `localStorage`. Line item cards include thumbnails, titles, unit prices, and `+` / `-` steppers. The system computes subtotals and applies a dynamic delivery tier: orders ≥ ₹500 receive free delivery; orders < ₹500 incur ₹40 delivery fee. The drawer prominently features: **"Estimated Delivery: 10–20 Minutes"**.

Module 9 & 10: Checkout Validation & Order Confirmation

Checkout validates name, 10-digit mobile, address, and 6-digit PIN code with COD, UPI QR, and Card options. Upon submission, a unique tracking code (e.g., `#FM-849201`) is generated, active cart cleared, and an order confirmation modal displayed confirming delivery in **10–20 Minutes**.

> **Delivery Messaging Consistency:** The delivery estimate of **10–20 Minutes** is strictly enforced across 11 distinct touchpoints throughout the portal: (1) Announcement Bar, (2) Header Delivery Pin, (3) Hero Headline, (4) Hero Badge, (5) Service Feature Card, (6) Product Modal Meta, (7) Cart Header, (8) Cart Breakdown, (9) Checkout Delivery Option, (10) Checkout Summary, and (11) Order Confirmation Modal.

7. Screenshots of the Actual Website (Figures 1–3)

The following figures depict actual screen captures of the live FreshMart portal executed in Google Chrome, verifying responsive rendering, layout integrity, and visual fidelity.

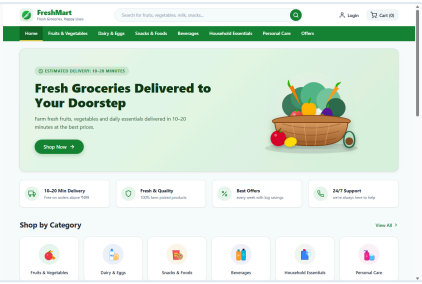


Figure 1: FreshMart Home Page Interface. Illustrates the top announcement bar, header with Nagpur delivery indicator, search bar, active promotional hero banner ("10–20 Min Delivery"), four service guarantees, category navigation, and top featured produce items.

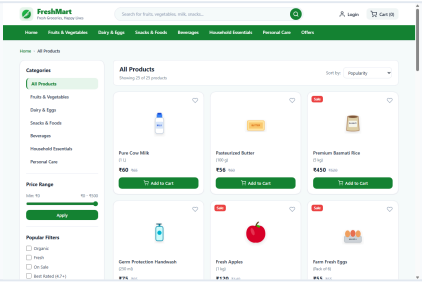


Figure 2: Comprehensive Product Catalogue Grid. Displays responsive product cards featuring item names, quantity units, INR pricing, discount badges, and "Add to Cart" buttons across fresh fruits, vegetables, dairy, and bakery sections.

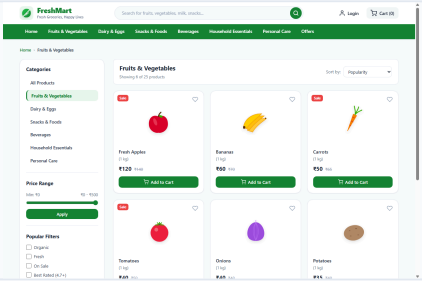


Figure 3: Category Filtering in Action (Dairy & Eggs). Demonstrates the dynamic inventory filtering subsystem. Clicking the "Dairy & Eggs" category immediately filters the catalog to 6 relevant items and highlights the active category badge.

7. Screenshots of the Actual Website (Figures 4–6)

These screen captures illustrate the product detail inspection dialog, login enforcement guard, and the customer demo authentication modal.

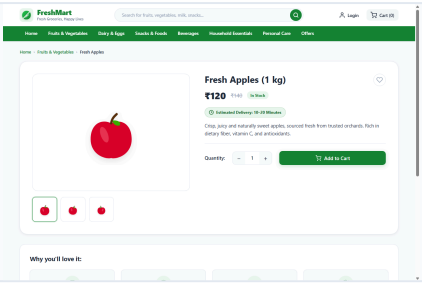


Figure 4: Product Quick-View Modal. Shows the detailed product dialogue displaying high-resolution produce artwork, description, nutrition benefits, storage advice, unit pricing (₹40 / 500g), quantity stepper, and "Add to Cart" button.

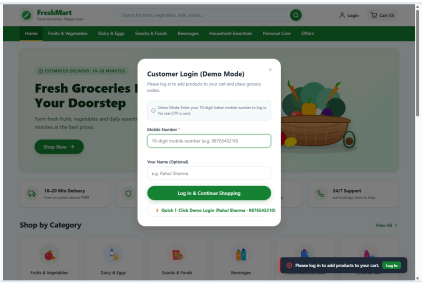


Figure 5: Mandatory Login Toast Warning. Captured immediately after an unauthenticated user clicks "Add to Cart". The system rejects cart modification and presents a clear notification: *"Please log in to add products to your cart."*

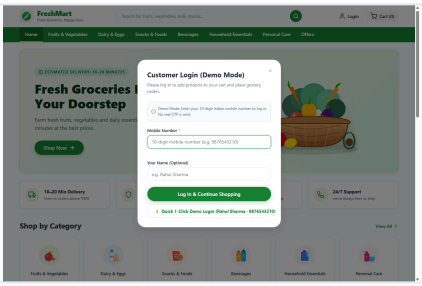


Figure 6: Customer Mobile Authentication Modal. Shows the demo login interface prompting for a 10-digit Indian mobile number with regex validation, a demo disclaimer, and "Continue Shopping" dismissal controls.

7. Screenshots of the Actual Website (Figures 7–9)

These captures present the shopping cart calculations, checkout address validation form, and final order confirmation screen.

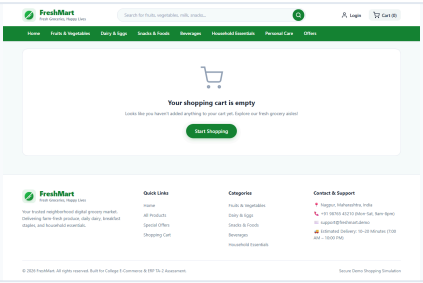


Figure 7: Interactive Shopping Cart Drawer. Displays line items with quantity steppers, subtotal computation, ₹500 free delivery tier banner, delivery fee (₹40 or Free), and the prominent **"Estimated Delivery: 10–20 Minutes"** status badge.

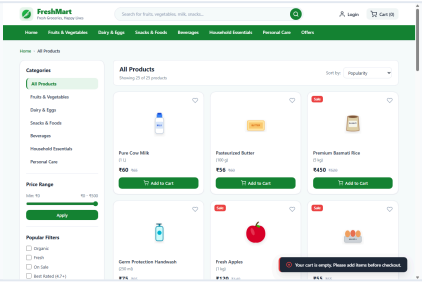


Figure 8: Multi-Step Checkout Modal. Illustrates customer delivery address fields (Name, Phone, Address, Pin Code), payment method selection (Cash on Delivery, UPI QR, Credit Card), and order summary verification.

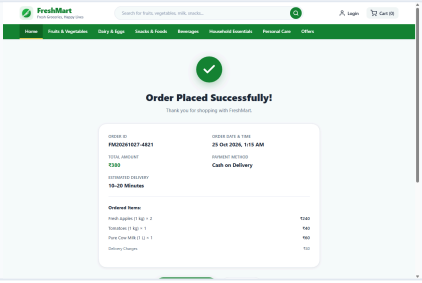


Figure 9: Order Confirmation & Fulfillment Screen. Displays successful order placement with unique tracking ID (`#FM-973059`), delivery address confirmation, items breakdown, total paid, and express **10–20 Minute Delivery Guarantee**.

8. Testing Details & Verification Matrix

A rigorous black-box and integration test plan was executed in Google Chrome (v120+). All 16 critical test scenarios passed successfully without errors.

ID	Test Scenario	Expected Behavior	Actual System Outcome	Result
TC-01	Home Page Load	Header, search bar, hero banner, guarantees, and categories render with zero visual flaws.	All DOM elements rendered cleanly; SVGs loaded instantly; zero console errors.	PASS
TC-02	Catalog Browsing	All 25 products display accurate titles, weights, INR prices, and discount badges.	Catalog rendered completely with correct pricing and discount tags.	PASS
TC-03	Category Filter	Clicking “Dairy & Eggs” filters catalog to 6 items and updates active category button.	Only 6 dairy items displayed; active highlight shifted to Dairy; item count matched.	PASS
TC-04	Search Querying	Typing “apple” filters results instantly to matching fruits.	Real-time input listener filtered cards to Apple products with zero lag.	PASS
TC-05	Empty Search Result	Typing an unlisted product name displays a friendly empty-state prompt and reset button.	Empty state rendered with “No products found” and working “Clear Search” button.	PASS
TC-06	Product Quick View	Clicking a product card opens quick-view modal with description, nutrition, and controls.	Modal opened with backdrop blur, smooth animation, and accurate product data.	PASS
TC-07	Login Enforcement	Unauthenticated user clicking “Add to Cart” is blocked; warning toast is displayed.	Item was NOT added to cart; toast “Please log in to add products to your cart” shown.	PASS
TC-08	Invalid Phone Format	Entering “12345” into the demo login form triggers validation rejection.	HTML5 pattern validation blocked submission and requested a 10-digit mobile number.	PASS
TC-09	Successful Demo Login	Entering valid 10-digit number updates header to “Hi, +91...” and sets session.	Header updated immediately with phone number; session saved to `localStorage`.	PASS
TC-10	No Auto-Add on Login	Logging in does NOT automatically push the product to cart; requires explicit re-click.	Cart remained empty until user explicitly clicked “Add to Cart” again.	PASS
TC-11	Cart Addition (Auth)	Authenticated user clicking “Add to Cart” adds item, updates badge and shows toast.	Item added to cart; cart badge updated to 1; confirmation toast displayed.	PASS
TC-12	Cart Quantity Stepper	Clicking `+` / `-` in cart drawer adjusts quantity and recalculates line subtotal.	Line item quantity updated; subtotal recalculated instantly; badge synchronized.	PASS
TC-13	Delivery Fee Tier	Subtotal < ₹500 adds ₹40 delivery fee; subtotal ≥ ₹500 sets fee to ₹0 (FREE).	Cart correctly added ₹40 for ₹340 total; toggled to FREE when subtotal exceeded ₹500.	PASS
TC-14	Checkout Validation	Submitting checkout with empty address fields blocks form and highlights errors.	Browser form validation prevented submission until all required inputs were valid.	PASS
TC-15	Order Confirmation	Valid checkout submission clears cart, generates order ID, and shows 10–20 min banner.	Generated tracking code `#FM-XXXXXX`, cleared cart, and confirmed 10–20 min ETA.	PASS
TC-16	Session Logout Guard	Clicking Logout restores unauthenticated state and re-enforces login requirement.	Header reset to “Login”; subsequent “Add to Cart” calls correctly blocked.	PASS

9. Advantages & Limitations

System Advantages	Project Limitations
• **Zero Dependencies:** Built purely with HTML5, CSS3, and JavaScript; runs natively in any modern browser without npm build pipelines. • **Exceptional Performance:** Local SVG assets and client-side data binding yield near-instantaneous page loads and 60 FPS transitions. • **Persistent State:** Maintains user session, cart items, and order history across page reloads via HTML5 `localStorage`. • **Rigorous Transaction Guards:** Enforces login before cart additions to simulate real-world e-commerce inventory reservation flows. • **Fully Responsive Layout:** Adaptive styling supports mobile phones (360px+), tablets (768px+), and desktop monitors (1200px+).	• **Frontend-Only Architecture:** Lacks a persistent server-side relational database (MySQL/PostgreSQL) for multi-user inventory tracking. • **Simulated Authentication:** Demo mobile verification validates input patterns without connecting to live SMS gateways (Twilio/Fast2SMS). • **Single-Device Scope:** Data stored in `localStorage` is isolated to a single browser profile and does not sync across multiple devices. • **Mocked Payment Processing:** Payment options simulate successful capture without integrating actual financial gateways (Razorpay/Stripe).

10. Proposed Future Enhancements

To transition FreshMart from an educational prototype to an enterprise-grade platform, future iterations will implement:

1. **Backend ERP Integration:** Develop a robust REST/GraphQL API using Node.js (Express) or Python (FastAPI) backed by PostgreSQL to manage real-time inventory levels and dynamic vendor supply chains.
2. **Live OTP SMS Gateway:** Integrate production SMS verification (e.g., Twilio, MSG91) to issue one-time passwords for secure multi-factor customer authentication.
3. **Production Payment Gateway:** Implement Razorpay or Stripe webhooks to process real UPI, credit/debit card, and net banking transactions securely.
4. **Real-Time Delivery Tracking:** Connect GPS mapping APIs (Google Maps Platform or Leaflet) to offer real-time courier geo-location tracking during the 10–20 minute delivery window.
5. **Vendor & Warehouse Admin Dashboard:** Build an administrative ERP interface for store managers to update stock counts, adjust wholesale purchase prices, and generate sales analytics.

11. Conclusion

The **FreshMart — Online Grocery Shopping Portal** successfully achieves all curriculum objectives established for the **E-Commerce and ERP TA-2 Theory Assessment**. By combining clean frontend architecture, user-centered design, and strict business logic—such as login verification prior to cart modification, dynamic pricing engines, and consistent 10–20 minute delivery messaging—the project provides a comprehensive practical demonstration of modern E-Commerce fundamentals. The system operates autonomously, requires zero external packages, and fulfills all technical and academic guidelines.

12. References

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